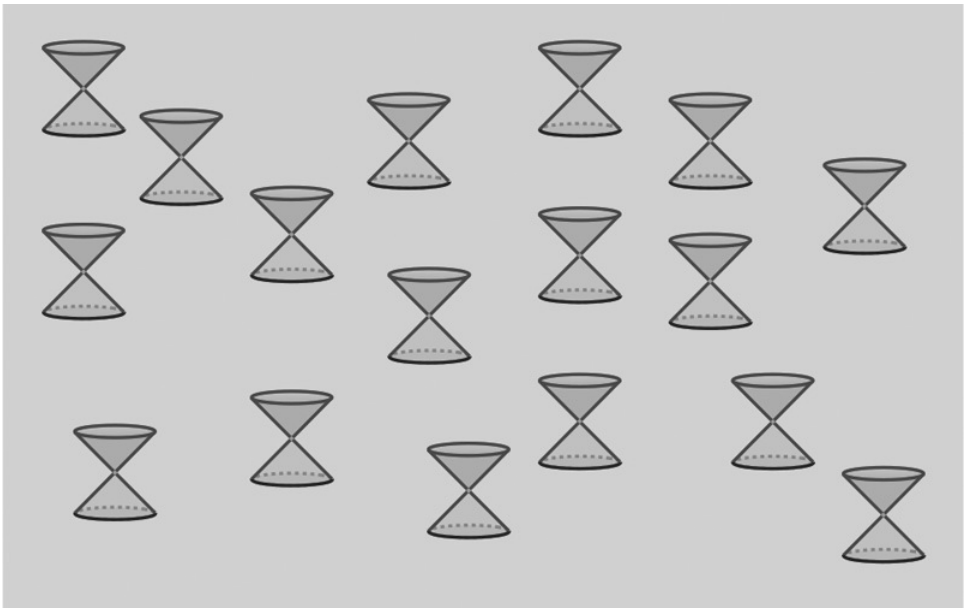
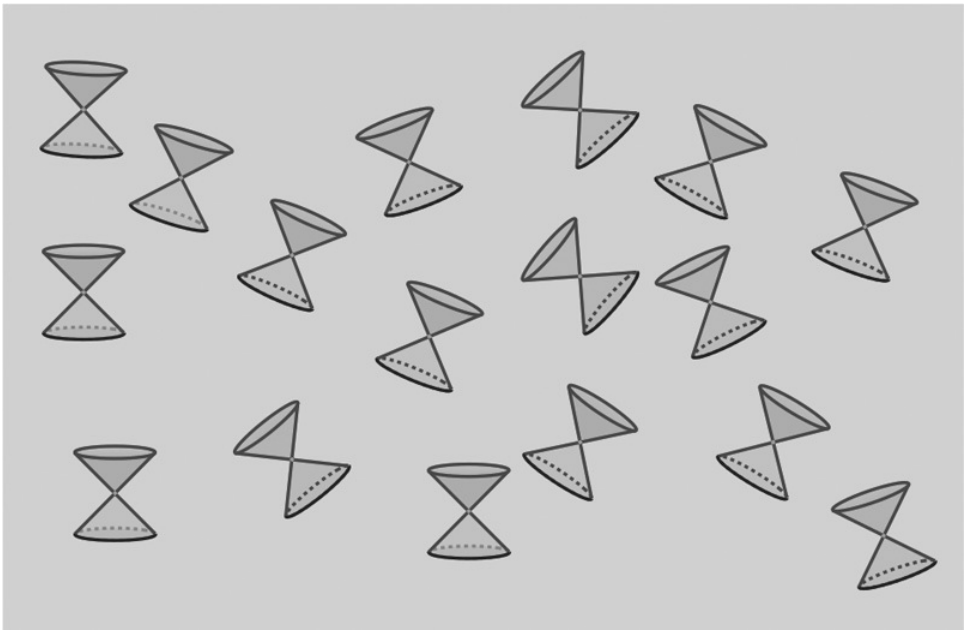


This is the structure of spacetime that Einstein understood when he was twenty-five years old.

Ten years later, he comes to understand that the speed at which time flows changes from place to place. It follows that spacetime does not really have the order outlined above but can be distorted. It now looks rather more like this:



When a gravitational wave passes, for example, the small light cones oscillate together from right to left, like ears of wheat blown by the wind.

The structure of the cones can even be such that, advancing always toward the future, one can return to the same point in spacetime, like this: